

PSDA · frame + one-variable data · ANSWERS

41 items · two PSDA families only · untimed

1–11 Family A · official · frame / generalize / MoE / experiment

12–25 Family B · official · frequency / box plot / two data sets

26–35 Family A · repo originals (Sets 06 / 07 + desk)

36–41 Family B · repo originals (Sets 03 / 04 / 06)

Official pages are the Hard exclude-active bank. Originals are not College Board stems. Do not re-sit Bluebook 4 / 5 / 6 / 8 / 9 / 11.

Instructor only. Official keys pages include the rationale.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Hard

Question

Texting behavior	Talks on cell phone daily	Does not talk on cell phone daily	Total
Light	110	146	256
Medium	139	164	303
Heavy	166	74	240
Total	415	384	799

In a study of cell phone use, 799 randomly selected US teens were asked how often they talked on a cell phone and about their texting behavior. The data are summarized in the table above. Based on the data from the study, an estimate of the percent of US teens who are heavy texters is 30% and the associated margin of error is 3%. Which of the following is a correct statement based on the given margin of error?

Answer

- A. Approximately 3% of the teens in the study who are classified as heavy texters are not really heavy texters.
- B. It is not possible that the percent of all US teens who are heavy texters is less than 27%.
- C. The percent of all US teens who are heavy texters is 33%.
- D. It is doubtful that the percent of all US teens who are heavy texters is 35%.

Correct Answer: D

Rationale

Choice D is correct. The given margin of error of 3% indicates that the actual percent of all US teens who are heavy texters is likely within 3% of the estimate of 30%, or between 27% and 33%. Therefore, it is unlikely, or doubtful, that the percent of all US teens who are heavy texters would be 35%.

Choice A is incorrect. The margin of error doesn't provide any information about the accuracy of reporting in the study. Choice B is incorrect. Based on the estimate and given margin of error, it is unlikely that the percent of all US teens who are heavy texters would be less than 27%, but it is possible. Choice C is incorrect. While the percent of all US teens who are heavy texters is likely between 27% and 33%, any value within this interval is equally likely. We can't be certain that the value is exactly 33%.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard

Question

A sample of 40 fourth-grade students was selected at random from a certain school. The 40 students completed a survey about the morning announcements, and 32 thought the announcements were helpful. Which of the following is the largest population to which the results of the survey can be applied?

Answer

- A. The 40 students who were surveyed
- B. All fourth-grade students at the school
- C. All students at the school
- D. All fourth-grade students in the county in which the school is located

Correct Answer: B

Rationale

Choice B is correct. Selecting a sample of a reasonable size at random to use for a survey allows the results from that survey to be applied to the population from which the sample was selected, but not beyond this population. In this case, the population from which the sample was selected is all fourth-grade students at a certain school. Therefore, the results of the survey can be applied to all fourth-grade students at the school.

Choice A is incorrect. The results of the survey can be applied to the 40 students who were surveyed. However, this isn't the largest group to which the results of the survey can be applied. Choices C and D are incorrect. Since the sample was selected at random from among the fourth-grade students at a certain school, the results of the survey can't be applied to other students at the school or to other fourth-grade students who weren't represented in the survey results. Students in other grades in the school or other fourth-grade students in the country may feel differently about announcements than the fourth-grade students at the school.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Hard

Question

Sample	Percent in favor	Margin of error
A	52%	4.2%
B	48%	1.6%

The results of two random samples of votes for a proposition are shown above. The samples were selected from the same population, and the margins of error were calculated using the same method. Which of the following is the most appropriate reason that the margin of error for sample A is greater than the margin of error for sample B?

Answer

- A. Sample A had a smaller number of votes that could not be recorded.
- B. Sample A had a higher percent of favorable responses.
- C. Sample A had a larger sample size.
- D. Sample A had a smaller sample size.

Correct Answer: D

Rationale

Choice D is correct. Sample size is an appropriate reason for the margin of error to change. In general, a smaller sample size increases the margin of error because the sample may be less representative of the whole population.

Choice A is incorrect. The margin of error will depend on the size of the sample of recorded votes, not the number of votes that could not be recorded. In any case, the smaller number of votes that could not be recorded for sample A would tend to decrease, not increase, the comparative size of the margin of error. Choice B is incorrect. Since the percent in favor for sample A is the same distance from 50% as the percent in favor for sample B, the percent of favorable responses doesn't affect the comparative size of the margin of error for the two samples. Choice C is incorrect. If sample A had a larger margin of error than sample B, then sample A would tend to be less representative of the population. Therefore, sample A is not likely to have a larger sample size.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard

Question

A trivia tournament organizer wanted to study the relationship between the number of points a team scores in a trivia round and the number of hours that a team practices each week. For the study, the organizer selected **55** teams at random from all trivia teams in a certain tournament. The table displays the information for the **40** teams in the sample that practiced for at least **3** hours per week.

Hours practiced	Number of points per round		
	6 to 13 points	14 or more points	Total
3 to 5 hours	6	4	10
More than 5 hours	4	26	30
Total	10	30	40

Which of the following is the largest population to which the results of the study can be generalized?

Answer

- A. All trivia teams in the tournament that scored **14** or more points in the round
- B. The **55** trivia teams in the sample
- C. The **40** trivia teams in the sample that practiced for at least **3** hours per week
- D. All trivia teams in the tournament

Correct Answer: D

Rationale

Choice D is correct. It's given that the organizer selected **55** teams at random from all trivia teams in the tournament. A table is also given displaying the information for the **40** teams in the sample that practiced for at least **3** hours per week. Selecting a sample of a reasonable size at random to use for a survey allows the results from that survey to be applied to the population from which the sample was selected, but not beyond this population. Thus, only the sampling method information is necessary to determine the largest population to which the results of the study can be generalized. Since the organizer selected the sample at random from all trivia teams in the tournament, the largest population to which the results of the study can be generalized is all trivia teams in the tournament.

Choice A is incorrect. The sample was selected at random from all trivia teams in the tournament, not just from the teams that scored an average of **14** or more points per round.

Choice B is incorrect. If a study uses a sample selected at random from a population, the results of the study can be generalized to the population, not just the sample.

Choice C is incorrect. If a study uses a sample selected at random from a population, the results of the study can be generalized to the population, not just a subset of the sample.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard

Question

A psychologist designed and conducted a study to determine whether playing a certain educational game increases middle school students' accuracy in adding fractions. For the study, the psychologist chose a random sample of 35 students from all of the students at one of the middle schools in a large city. The psychologist found that students who played the game showed significant improvement in accuracy when adding fractions. What is the largest group to which the results of the study can be generalized?

Answer

- A. The 35 students in the sample
- B. All students at the school
- C. All middle school students in the city
- D. All students in the city

Correct Answer: B

Rationale

Choice B is correct. The largest group to which the results of a study can be generalized is the population from which the random sample was chosen. In this case, the psychologist chose a random sample from all students at one particular middle school. Therefore, the largest group to which the results can be generalized is all the students at the school.

Choice A is incorrect because this isn't the largest group the results can be generalized to. Choices C and D are incorrect because these groups are larger than the population from which the random sample was chosen. Therefore, the sample isn't representative of these groups.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard

Question

A statistician conducted a study to investigate the relationship between the average number of minutes piano students practiced each week and the percentage of correct notes the students played during a recital. For the study, the statistician selected a sample of **85** piano students at random from all the piano students in a certain region who were in their second year with their conservatory. The table shows the information for **75** piano students in this sample who practiced at least **50** minutes on average each week.

	Percentage of correct notes the students played during a recital			
Average number of minutes practiced each week	Less than 50%	50% to 80%	Greater than 80%	Total
50 to 150	7	28	5	40
151 or more	3	11	21	35
Total	10	39	26	75

Which of the following is the largest population to which the results of the study can be generalized?

Answer

- A. All piano students who were in their second year with their conservatory
- B. All piano students in the region
- C. All piano students in the region who were in their second year with their conservatory
- D. The **75** piano students in the sample who practiced at least **50** minutes on average each week

Correct Answer: C

Rationale

Choice C is correct. It's given that the statistician selected a sample of **85** piano students at random from all piano students in a certain region who were in their second year with their conservatory. It's also given that the table shows the information for **75** piano students in the sample who practiced at least **50** minutes on average each week. Selecting a sample at random from a population allows the results of the study to be generalized to that population, but not beyond it. Therefore, the largest population to which the results of the study can be generalized is the population of all piano students in the region who were in their second year with their conservatory.

Choice A is incorrect. The results can't be generalized to all piano students who were in their second year with their conservatory since only students in a certain region were sampled.

Choice B is incorrect. The results can't be generalized to all piano students in the region since only students in their second year with their conservatory were sampled.

Choice D is incorrect. The results of the study can be generalized beyond the sample itself to the population from which it was drawn.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard

Question

Near the end of a US cable news show, the host invited viewers to respond to a poll on the show’s website that asked, “Do you support the new federal policy discussed during the show?” At the end of the show, the host reported that 28% responded “Yes,” and 70% responded “No.” Which of the following best explains why the results are unlikely to represent the sentiments of the population of the United States?

Answer

- A. The percentages do not add up to 100%, so any possible conclusions from the poll are invalid.
- B. Those who responded to the poll were not a random sample of the population of the United States.
- C. There were not 50% “Yes” responses and 50% “No” responses.
- D. The show did not allow viewers enough time to respond to the poll.

Correct Answer: B

Rationale

Choice B is correct. In order for the poll results from a sample of a population to represent the entire population, the sample must be representative of the population. A sample that is randomly selected from a population is more likely than a sample of the type described to represent the population. In this case, the people who responded were people with access to cable television and websites, which aren’t accessible to the entire population. Moreover, the people who responded also chose to watch the show and respond to the poll. The people who made these choices aren’t representative of the entire population of the United States because they were not a random sample of the population of the United States.

Choices A, C, and D are incorrect because they present reasons unrelated to whether the sample is representative of the population of the United States.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard

Question

To determine the mean number of children per household in a community, Tabitha surveyed 20 families at a playground. For the 20 families surveyed, the mean number of children per household was 2.4. Which of the following statements must be true?

Answer

- A. The mean number of children per household in the community is 2.4.
- B. A determination about the mean number of children per household in the community should not be made because the sample size is too small.
- C. The sampling method is flawed and may produce a biased estimate of the mean number of children per household in the community.
- D. The sampling method is not flawed and is likely to produce an unbiased estimate of the mean number of children per household in the community.

Correct Answer: C

Rationale

Choice C is correct. In order to use a sample mean to estimate the mean for a population, the sample must be representative of the population (for example, a simple random sample). In this case, Tabitha surveyed 20 families in a playground. Families in the playground are more likely to have children than other households in the community. Therefore, the sample isn't representative of the population. Hence, the sampling method is flawed and may produce a biased estimate.

Choices A and D are incorrect because they incorrectly assume the sampling method is unbiased. Choice B is incorrect because a sample of size 20 could be large enough to make an estimate if the sample had been representative of all the families in the community.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Hard

Question

A researcher surveyed a random sample of **1,020** people from a certain city to estimate the percentage of people who support a recently announced proposal. From the survey, the researcher estimated that **86%** of people from the city support this proposal, with an associated margin of error of **2.13%**. The researcher repeated the survey with a random sample of people from the city that was double the original sample size. Assuming the margins of error were calculated in the same way, which of the following is true about the margin of error associated with the estimate from the larger sample?

Answer

- A. The margin of error is between **2.13%** and **3.13%**.
- B. The margin of error is between **4.13%** and **5.13%**.
- C. The margin of error is greater than **5.13%**.
- D. The margin of error is less than **2.13%**.

Correct Answer: D

Rationale

Choice D is correct. It's given that from the initial survey, the margin of error associated with the estimate was **2.13%**. A larger sample size generally leads to a smaller margin of error. Therefore, assuming the margins of error were calculated in the same way, the margin of error associated with the estimate from the larger sample is less than **2.13%**.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Inference from sample statistics and margin of error	Hard

Question

In State X, Mr. Camp’s eighth-grade class consisting of 26 students was surveyed and 34.6 percent of the students reported that they had at least two siblings. The average eighth-grade class size in the state is 26. If the students in Mr. Camp’s class are representative of students in the state’s eighth-grade classes and there are 1,800 eighth-grade classes in the state, which of the following best estimates the number of eighth-grade students in the state who have fewer than two siblings?

Answer

- A. 16,200
- B. 23,400
- C. 30,600
- D. 46,800

Correct Answer: C

Rationale

Choice C is correct. It is given that 34.6% of 26 students in Mr. Camp’s class reported that they had at least two siblings. Since 34.6% of 26 is 8.996, there must have been 9 students in the class who reported having at least two siblings and 17 students who reported that they had fewer than two siblings. It is also given that the average eighth-grade class size in the state is 26 and that Mr. Camp’s class is representative of all eighth-grade classes in the state. This means that in each eighth-grade class in the state there are about 17 students who have fewer than two siblings. Therefore, the best estimate of the number of eighth-grade students in the state who have fewer than two siblings is $17 \times (\text{number of eighth-grade classes in the state})$, or $17 \times 1,800 = 30,600$.

Choice A is incorrect because 16,200 is the best estimate for the number of eighth-grade students in the state who have at least, not fewer than, two siblings. Choice B is incorrect because 23,400 is half of the estimated total number of eighth-grade students in the state; however, since the students in Mr. Camp’s class are representative of students in the eighth-grade classes in the state and more than half of the students in Mr. Camp’s class have fewer than two siblings, more than half of the students in each eighth-grade class in the state have fewer than two siblings, too. Choice D is incorrect because 46,800 is the estimated total number of eighth-grade students in the state.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Evaluating statistical claims: Observational studies and experiments	Hard

Question

For a baobab tree habitat in South Africa, a scientist randomly selected **50** baobab trees that were **17** years old and randomly assigned them to two groups. Each group was subjected to a different watering pattern for **2** consecutive years to observe whether the watering pattern affects the trees’ growth rate. Based on the design of the study, what is the largest group to which these results can be applied?

Answer

- A. All the **50** baobab trees that were selected in this habitat
- B. All the baobab trees that were **19** years old in this habitat
- C. All the baobab trees that were **17** years old in South Africa
- D. All the baobab trees that were **17** years old in this habitat

Correct Answer: D

Rationale

Choice D is correct. When a study uses a randomly selected sample, the largest group to which the results of the study can be applied is the population from which the sample was selected. It's given that the scientist randomly selected the trees from the baobab trees in a certain habitat that were **17** years old. Therefore, the largest group to which the results of this study can be applied is all the baobab trees that were **17** years old in this habitat.

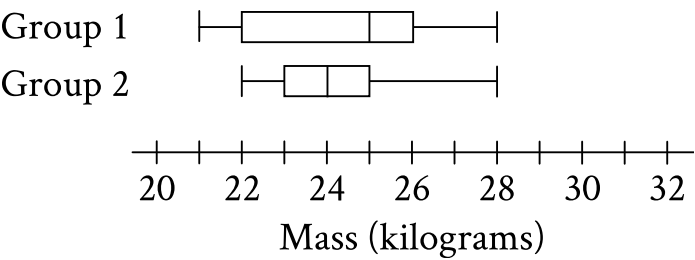
Choice A is incorrect. Since the sample was randomly selected from a population, the results can be applied to a larger group than the sample.

Choice B is incorrect. The sample was selected from a population of baobab trees that were **17** years old, not **19** years old.

Choice C is incorrect. The sample was selected from a certain tree habitat in South Africa, not from all the baobab trees that were **17** years old in South Africa.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question



The box plots summarize the masses, in kilograms, of two groups of gazelles. Based on the box plots, which of the following statements must be true?

Answer

- A. The mean mass of group 1 is greater than the mean mass of group 2.
- B. The mean mass of group 1 is less than the mean mass of group 2.
- C. The median mass of group 1 is greater than the median mass of group 2.
- D. The median mass of group 1 is less than the median mass of group 2.

Correct Answer: C

Rationale

Choice C is correct. The median of a data set represented in a box plot is represented by the vertical line within the box. It follows that the median mass of the gazelles in group 1 is 25 kilograms, and the median mass of the gazelles in group 2 is 24 kilograms. Since 25 kilograms is greater than 24 kilograms, the median mass of group 1 is greater than the median mass of group 2.

Choice A is incorrect. The mean mass of each of the two groups cannot be determined from the box plots.

Choice B is incorrect. The mean mass of each of the two groups cannot be determined from the box plots.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

For data set A, the table summarizes the distribution of the number of deliveries received by an office each day during a period of **11** days.

Deliveries	Days
0	2
3	2
4	2
5	2
6	1
7	1
13	1

The data value **13** was recorded in error and is removed from data set A to create data set B, which consists of the remaining **10** data values. Which statement best compares the median of data set A and the median of data set B?

Answer

- A. The median of data set B is less than the median of data set A.
- B. The median of data set B is greater than the median of data set A.
- C. The median of data set B is equal to the median of data set A.
- D. There is not enough information to compare the medians of the two data sets.

Correct Answer: C

Rationale

Choice C is correct. It’s given that for data set A, the table summarizes the number of deliveries received by an office each day over a period of **11** days. If a data set contains an odd number of data values, the median is represented by the middle data value in the list when the data values are listed in ascending or descending order. Since data set A contains **11** data values in ascending order, the median of data set A is the **6th** value. Based on the number of days for each value shown in the table, the **6th** value is **4**, so the median of data set A is **4**. It’s also given that the data value **13** was recorded in error and is removed from data set A to create data set B. It follows that data set B contains **10** data values in ascending order. If a data set contains an even number of data values, the median is between the two middle data values when the values are listed in ascending or descending order. It follows that the median of data set B is between the **5th** and **6th** values. Based on the number of days for each value shown in the table, the **5th** and **6th** values are both **4**, so the median of data set B is **4**. Therefore, the median of data set B is equal to the median of data set A.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Data set A consists of **10** positive integers less than **60**. The list shown gives **9** of the integers from data set A.

43, 45, 44, 43, 38, 39, 40, 46, 40

The mean of these **9** integers is **42**. If the mean of data set A is an integer that is greater than **42**, what is the value of the largest integer from data set A?

Correct Answer: 52

Rationale

The correct answer is **52**. The mean of a data set is calculated by dividing the sum of the data values by the number of values. It's given that data set A consists of **10** values, **9** of which are shown. Let x represent the **10th** data value in data set A, which isn't shown. The mean of data set A can be found using the expression $\frac{43+45+44+43+38+39+40+46+40+x}{10}$, or $\frac{378+x}{10}$. It's given that the mean of the **9** values shown is **42** and that the mean of all **10** numbers is greater than **42**. Consequently, the **10th** data value, x , is larger than **42**. It's also given that the data values in data set A are positive integers less than **60**. Thus, $42 < x < 60$. Finally, it's given that the mean of data set A is an integer. This means that the sum of the **10** data values, $378 + x$, is divisible by **10**. Thus, $378 + x$ must have a ones digit of **0**. It follows that x must have a ones digit of **2**. Since $42 < x < 60$ and x has a ones digit of **2**, the only possible value of x is **52**. Since **52** is larger than any of the integers shown, the largest integer from data set A is **52**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Data Set Q

Value	Frequency
a	110
b	30
c	0

Data Set R

Value	Frequency
a	0
b	3
c	11

Data sets Q and R are summarized in the tables shown, where a , b , and c are positive, consecutive integers and $a < b < c$. Which statement comparing the means of the data sets is true?

Answer

- A. The mean of data set R is $\frac{2(11)}{14}$ less than the mean of data set Q.
- B. The mean of data set R is $\frac{2(11)}{14}$ greater than the mean of data set Q.
- C. The mean of data set Q is 10 times the mean of data set R.
- D. The means of the two data sets are equal.

Correct Answer: B

Rationale

Choice B is correct. The mean of a data set is the sum of the values of the data set divided by the total number of values in the data set. When a data set is represented in a frequency table, the sum of the values in the data set is the sum of the products of each value and its frequency. For data set Q, the sum of the products of each value and its frequency is $110a + 30b + 0c$, or $110a + 30b$, and the number of values is $110 + 30 + 0$, or 140. Therefore, the mean of data set Q is $\frac{110a+30b}{140}$. For data set R, the sum of the products of each value and its frequency is $0a + 3b + 11c$, or $3b + 11c$, and the number of values is $0 + 3 + 11$, or 14. Therefore, the mean of data set R is $\frac{3b+11c}{14}$. Since a , b , and c are positive, consecutive integers and $a < b < c$, it follows that $b = a + 1$ and $c = a + 2$. Substituting $a + 1$ for b in the mean of data set Q yields $\frac{110a+30(a+1)}{140}$, or $\frac{110a+30a+30}{140}$, which is equivalent to $\frac{140a+30}{140}$, or $a + \frac{3}{14}$. Substituting $a + 1$ for b and $a + 2$ for c in the mean of data set R yields $\frac{3(a+1)+11(a+2)}{14}$, or $\frac{3a+3+11a+22}{14}$, which is equivalent to $\frac{14a+25}{14}$, or $a + \frac{3}{14} + \frac{2(11)}{14}$. Subtracting the mean of data set Q from the mean of data set R yields $\left(a + \frac{3}{14} + \frac{2(11)}{14}\right) - \left(a + \frac{3}{14}\right)$, or $\frac{2(11)}{14}$. Therefore, the mean of data set R is $\frac{2(11)}{14}$ greater than the mean of data set Q.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Value	Data set A frequency	Data set B frequency
30	2	9
34	4	7
38	5	5
42	7	4
46	9	2

Data set A and data set B each consist of **27** values. The table shows the frequencies of the values for each data set. Which of the following statements best compares the means of the two data sets?

Answer

- A. The mean of data set A is greater than the mean of data set B.
- B. The mean of data set A is less than the mean of data set B.
- C. The mean of data set A is equal to the mean of data set B.
- D. There is not enough information to compare the means of the data sets.

Correct Answer: A

Rationale

Choice A is correct. The mean value of a data set is the sum of the values of the data set divided by the number of values in the data set. When a data set is represented in a frequency table, the sum of the values in the data set is the sum of the products of each value and its frequency. For data set A, the sum of products of each value and its frequency is $30(2) + 34(4) + 38(5) + 42(7) + 46(9)$, or **1,094**. It's given that there are **27** values in data set A. Therefore, the mean of data set A is $\frac{1,094}{27}$, or approximately **40.52**. Similarly, the mean of data B is $\frac{958}{27}$, or approximately **35.48**. Therefore, the mean of data set A is greater than the mean of data set B.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

A data set of 27 different numbers has a mean of 33 and a median of 33. A new data set is created by adding 7 to each number in the original data set that is greater than the median and subtracting 7 from each number in the original data set that is less than the median. Which of the following measures does NOT have the same value in both the original and new data sets?

Answer

- A. Median
- B. Mean
- C. Sum of the numbers
- D. Standard deviation

Correct Answer: D

Rationale

Choice D is correct. When a data set has an odd number of elements, the median can be found by ordering the values from least to greatest and determining the middle value. Out of the 27 different numbers in this data set, 13 numbers are below the median, one number is exactly 33, and 13 numbers are above the median. When 7 is subtracted from each number below the median and added to each number above the median, the data spread out from the median. Since the median of this data set, 33, is equivalent to the mean of the data set, the data also spread out from the mean. Since standard deviation is a measure of how spread out the data are from the mean, a greater spread from the mean indicates an increased standard deviation.

Choice A is incorrect. All the numbers less than the median decrease and all the numbers greater than the median increase, but the median itself doesn't change. Choices B and C are incorrect. The mean of a data set is found by dividing the sum of the values by the number of values. The net change from subtracting 7 from 13 numbers and adding 7 to 13 numbers is zero. Therefore, neither the mean nor the sum of the numbers changes.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Value	Frequency
1	a
2	2a
3	3a
4	2a
5	a

The frequency distribution above summarizes a set of data, where a is a positive integer. How much greater is the mean of the set of data than the median?

Answer

- A. 0
- B. 1
- C. 2
- D. 3

Correct Answer: A

Rationale

Choice A is correct. Since the frequencies of values less than the middle value, 3, are the same as the frequencies of the values greater than 3, the set of data has a symmetric distribution. When a set of data has a symmetric distribution, the mean and median values are equal. Therefore, the mean is 0 greater than the median.

Choices B, C, and D are incorrect and may result from misinterpreting the set of data.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

The mean score of 8 players in a basketball game was 14.5 points. If the highest individual score is removed, the mean score of the remaining 7 players becomes 12 points. What was the highest score?

Answer

- A. 20
- B. 24
- C. 32
- D. 36

Correct Answer: C

Rationale

Choice C is correct. If the mean score of 8 players is 14.5, then the total of all 8 scores is $14.5 \times 8 = 116$. If the mean of 7 scores is 12, then the total of all 7 scores is $12 \times 7 = 84$. Since the set of 7 scores was made by removing the highest score of the set of 8 scores, then the difference between the total of all 8 scores and the total of all 7 scores is equal to the removed score: $116 - 84 = 32$.

Choice A is incorrect because if 20 is removed from the group of 8 scores, then the mean score of the remaining 7 players is $\frac{(14.5 \times 8) - 20}{7}$ is approximately 13.71, not 12. Choice B is incorrect because if 24 is removed from the group of 8 scores, then the mean score of the remaining 7 players is $\frac{(14.5 \times 8) - 24}{7}$ is approximately 13.14, not 12. Choice D is incorrect because if 36 is removed from the group of 8 scores, then the mean score of the remaining 7 players is $\frac{(14.5 \times 8) - 36}{7}$ or approximately 11.43, not 12.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

The mean amount of time that the 20 employees of a construction company have worked for the company is 6.7 years. After one of the employees leaves the company, the mean amount of time that the remaining employees have worked for the company is reduced to 6.25 years. How many years did the employee who left the company work for the company?

Answer

- A. 0.45
- B. 2.30
- C. 9.00
- D. 15.25

Correct Answer: D

Rationale

Choice D is correct. The mean amount of time that the 20 employees worked for the company is 6.7 years. This means that the total number of years all 20 employees worked for the company is $(6.7)(20) = 134$ years. After the employee left, the mean amount of time that the remaining 19 employees worked for the company is 6.25 years. Therefore, the total number of years all 19 employees worked for the company is $(6.25)(19) = 118.75$ years. It follows that the number of years that the employee who left had worked for the company is $134 - 118.75 = 15.25$ years.

Choice A is incorrect; this is the change in the mean, which isn't the same as the amount of time worked by the employee who left. Choice B is incorrect and likely results from making the assumption that there were still 20 employees, rather than 19, at the company after the employee left and then subtracting the original mean of 6.7 from that result. Choice C is incorrect and likely results from making the assumption that there were still 20 employees, rather than 19, at the company after the employee left.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

Data set A consists of the heights of **75** objects and has a mean of **25** meters. Data set B consists of the heights of **50** objects and has a mean of **65** meters. Data set C consists of the heights of the **125** objects from data sets A and B. What is the mean, in meters, of data set C?

Correct Answer: 41

Rationale

The correct answer is **41**. The mean of a data set is computed by dividing the sum of the values in the data set by the number of values in the data set. It's given that data set A consists of the heights of **75** objects and has a mean of **25** meters. This can be represented by the equation $\frac{x}{75} = 25$, where x represents the sum of the heights of the objects, in meters, in data set A. Multiplying both sides of this equation by **75** yields $x = 75(25)$, or $x = 1,875$ meters. Therefore, the sum of the heights of the objects in data set A is **1,875** meters. It's also given that data set B consists of the heights of **50** objects and has a mean of **65** meters. This can be represented by the equation $\frac{y}{50} = 65$, where y represents the sum of the heights of the objects, in meters, in data set B. Multiplying both sides of this equation by **50** yields $y = 50(65)$, or $y = 3,250$ meters. Therefore, the sum of the heights of the objects in data set B is **3,250** meters. Since it's given that data set C consists of the heights of the **125** objects from data sets A and B, it follows that the mean of data set C is the sum of the heights of the objects, in meters, in data sets A and B divided by the number of objects represented in data sets A and B, or $\frac{1,875+3,250}{125}$, which is equivalent to **41** meters. Therefore, the mean, in meters, of data set C is **41**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question
Percent of Residents Who Earned a Bachelor's Degree or Higher

State	Percent of residents
State A	21.9%
State B	27.9%
State C	25.9%
State D	19.5%
State E	30.1%
State F	36.4%
State G	35.5%

A survey was given to residents of all 50 states asking if they had earned a bachelor's degree or higher. The results from 7 of the states are given in the table above. The median percent of residents who earned a bachelor's degree or higher for all 50 states was 26.95%. What is the difference between the median percent of residents who earned a bachelor's degree or higher for these 7 states and the median for all 50 states?

- Answer**
- A. 0.05%
 - B. 0.95%
 - C. 1.22%
 - D. 7.45%

Correct Answer: B

Rationale
Choice B is correct. The median of a set of numbers is the middle value of the set values when ordered from least to greatest. If the percents in the table are ordered from least to greatest, the middle value is 27.9%. The difference between 27.9% and 26.95% is 0.95%.

Choice A is incorrect and may be the result of calculation errors or not finding the median of the data in the table correctly. Choice C is incorrect and may be the result of finding the mean instead of the median. Choice D is incorrect and may be the result of using the middle value of the unordered list.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

	Masses (kilograms)					
Andrew	2.4	2.5	3.6	3.1	2.5	2.7
Maria	x	3.1	2.7	2.9	3.3	2.8

Andrew and Maria each collected six rocks, and the masses of the rocks are shown in the table above. The mean of the masses of the rocks Maria collected is 0.1 kilogram greater than the mean of the masses of the rocks Andrew collected. What is the value of x ?

Rationale

The correct answer is 2.6. Since the mean of a set of numbers can be found by adding the numbers together and dividing by how many numbers there are in the set, the mean mass, in kilograms, of the rocks Andrew collected is $\frac{2.4+2.5+3.6+3.1+2.5+2.7}{6} = \frac{16.8}{6}$, or 2.8.

Since the mean mass of the rocks Maria collected is 0.1 kilogram greater than the mean mass of rocks Andrew collected, the mean mass of the rocks Maria collected is $2.8+0.1=2.9$ kilograms. The value of x can be found by writing an equation for finding the mean:

$\frac{x+3.1+2.7+2.9+3.3+2.8}{6} = 2.9$. Solving this equation gives $x = 2.6$. Note that 2.6 and 13/5 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

For a certain computer game, individuals receive an integer score that ranges from 2 through 10. The table below shows the frequency distribution of the scores of the 9 players in group A and the 11 players in group B.

Score	Score Frequencies	
	Group A	Group B
2	1	0
3	1	0
4	2	0
5	1	4
6	3	2
7	0	0
8	0	2
9	1	1
10	0	2
Total	9	11

The median of the scores for group B is how much greater than the median of the scores for group A?

Rationale

The correct answer is 1. When there are an odd number of values in a data set, the median of the data set is the middle number when the data values are ordered from least to greatest. The scores for group A, ordered from least to greatest, are 2, 3, 4, 4, 5, 6, 6, 6, and 9. The median of the scores for group A is therefore 5. The scores for group B, ordered from least to greatest, are 5, 5, 5, 5, 6, 6, 8, 8, 9, 10, and 10. The median of the scores for group B is therefore 6. The median score for group B is $6 - 5 = 1$ more than the median score for group A.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	One-variable data: Distributions and measures of center and spread	Hard

Question

$a, 26, 29, b, 31, 47, c$

For the given data set, the data values are listed in ascending order, where a , b , and c are constants. For this data set, the mean is **36**, the median is **29**, and the range is **72**. What is the value of c ?

Answer

- A. 54
- B. 72
- C. 81
- D. 98

Correct Answer: C

Rationale

Choice C is correct. It's given that the data set containing $a, 26, 29, b, 31, 47, c$ is listed in ascending order and has a mean of **36**, a median of **29**, and a range of **72**. Since there are **7** values in the given data set, it follows that the median is the value in the fourth position. It's given that b is in the fourth position in the given data set, so it follows that $b = 29$. Since the mean of the data set is **36**, it follows that $\frac{(a+26+29+29+31+47+c)}{7} = 36$. Multiplying both sides of this equation by **7** yields $a + 26 + 29 + 29 + 31 + 47 + c = 252$, which is equivalent to $a + c + 162 = 252$. Subtracting **162** from both sides of this equation yields $a + c = 90$. The range is the difference between the greatest and least values in a data set. Since c is the greatest value and a is the least value in the given data set, it follows that $c - a = 72$, or $-a + c = 72$. Adding the equations $a + c = 90$ and $-a + c = 72$ yields $2c = 162$. Dividing both sides of this equation by **2** yields $c = 81$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

26. Near the end of a city-council livestream, the host invited viewers to answer a poll on the show's website: "Do you support the new parking rule discussed tonight?" The host later reported that 31% answered "Yes" and 66% answered "No." Which of the following best explains why the results are unlikely to represent the residents of the city?

- A) The percentages do not add up to 100%, so any possible conclusions from the poll are invalid.
- B) Those who responded to the poll were not a random sample of the residents of the city.
- C) There were not 50% "Yes" responses and 50% "No" responses.
- D) The show did not allow viewers enough time to respond to the poll.

27. A botanist wants to know whether a new fertilizer changes tomato yield. She has 40 plants of the same variety. She assigns 20 plants at random to the new fertilizer and 20 to the old fertilizer, grows them in the same greenhouse, and compares the mean yield. Which statement is true?

- A) This is an observational study, so a difference in mean yield cannot be used to claim the fertilizer caused the change.
- B) This is an experiment with random assignment, so a difference in mean yield can be used to claim the fertilizer caused the change.
- C) The results can be generalized to all plants of every species, because 40 is a large sample.
- D) The study is biased because the two groups do not have the same number of plants.

28. A clinic director selected 80 patients at random from **all** patients at the clinic. The table shows only the 35 patients in that sample who had at least 4 visits last year.

	4 to 6 visits	7 or more visits	Total
Improved	12	9	21
Did not improve	8	6	14
Total	20	15	35

Which of the following is the largest population to which the results of the study can be generalized?

- A) All clinic patients who improved
- B) The 80 patients in the sample
- C) The 35 patients in the sample who had at least 4 visits
- D) All patients at the clinic

29. A school counselor notices that students who already eat breakfast in the cafeteria have higher math-quiz averages than students who do not. She did not assign anyone to eat or skip breakfast. Which statement is true?

- A) This is an experiment, so she can conclude that eating breakfast caused the higher averages.
- B) This is an observational study, so she cannot conclude that eating breakfast caused the higher averages.
- C) The finding can be generalized to all teenagers in the country, because a school is a random sample of the country.
- D) The two groups must have the same number of students, or the comparison is invalid.

30. Two random samples of votes on the same city measure were selected from the same population of voters. The margins of error were calculated the same way.

Sample	Percent in favor	Margin of error
A	54%	4.8%
B	50%	1.5%

Which of the following is the most appropriate reason that the margin of error for sample A is greater than the margin of error for sample B?

- A) Sample A had a higher percent of favorable responses.
- B) Sample A had a larger sample size.
- C) Sample A had a smaller sample size.
- D) Sample A had fewer ballots that could not be recorded.

31. A random sample of residents in a town shows that 38% support a new park, with a margin of error of 3%. Which conclusion is most appropriate?

- A) Exactly 38% of all residents in the town support the park.
- B) The true percent of all residents who support the park is likely between 35% and 41%.
- C) The park will be approved.
- D) The sample is biased because the result is not 50% to 50%.

32. Two random samples are taken from the same city. Sample P has 80 people. Sample Q has 800 people. The same method is used to compute each margin of error. Which sample is likely to have the **smaller** margin of error?

- A) Sample P
- B) Sample Q
- C) The two margins of error must be equal.
- D) It cannot be determined without the sample percents.

33. A random sample of voters shows 52% support, with a margin of error of 4%. Which conclusion is most appropriate?

- A) Exactly 52% of all voters support the candidate.
- B) The true support is likely between 48% and 56%.
- C) The candidate will win.
- D) The sample is biased because the result is not 50–50.

34. A researcher asks only students in the honors hallway whether they like a new schedule, then claims the **whole school** supports it. The main problem with this claim is:

- A) The mean is undefined.
- B) The sample is not representative of the school.
- C) The margin of error must be 0.
- D) This is a well-designed experiment.

35. A poll is 61% with margin of error 5%. Which statement is **not** justified?

- A) A reasonable interval for the true percent is 56% to 66%.
- B) More than half of the population might favor the idea.
- C) Exactly 61% of the whole population favors the idea.
- D) A larger random sample would likely give a smaller margin of error.

36. (SPR) The frequency table summarizes a data set.

Data value	Frequency
4	9
5	12
6	8
7	0
8	5
9	11

What is the maximum data value in the data set?

37. The frequency table summarizes a data set.

Data value	Frequency
10	2
20	20
30	8
40	1

What is the maximum frequency value in the data set?

- A) 20
- B) 40
- C) 400
- D) 61

38. (SPR) The frequency table summarizes a data set.

Data value	Frequency
2	4
5	15
8	7
11	2
14	6
17	0
18	3

What is the maximum data value in the data set?

39. Data set A is 8, 10, 10, 11, 15. Data set B is created by adding 7 to each value in data set A. Which statement is true?

- A) The median of B equals the median of A, and the range of B equals the range of A.
- B) The median of B equals the median of A, and the range of B is greater than the range of A.
- C) The median of B is greater than the median of A, and the range of B equals the range of A.
- D) The median of B is greater than the median of A, and the range of B is greater than the range of A.

40. (SPR) Two data sets of 25 integers each are summarized by the same frequencies.

Interval	Frequency
$0 \leq x < 10$	5
$10 \leq x < 20$	6
$20 \leq x < 30$	7
$30 \leq x < 40$	7

The first interval is the integers greater than or equal to 0 and less than 10, and so on. What is the **smallest** possible difference between the mean of data set A and the mean of data set B ?

41. (SPR) Two data sets of 23 integers each are summarized by the same frequencies.

Interval	Frequency
$20 \leq x < 30$	3
$30 \leq x < 40$	4
$40 \leq x < 50$	7
$50 \leq x < 60$	9

The first interval is the integers greater than or equal to 20 and less than 30, and so on. What is the **greatest** possible difference between the mean of data set A and the mean of data set B ?